

Error budget for the $K = 1$, $a = 4 \text{ } \mu\text{m}$ champion

Which of $|P\rangle$ decay, Rydberg decay, or same-species V_{cc} is the leak?

TriQG examples/Smaller_VDD_energy · a4_K1_error_budget.py · 2026-05-15

One-line answer: The two gates have **different** dominant errors. Re-running the champion drive parameters with each error channel switched off (`a4_K1_error_budget.py`) gives, for the basis-fidelity infidelity $1 - \overline{F}_{\text{basis}}$:

- **OR gate** ($T_{\text{tot}} = 244 \text{ ns}$): Rydberg decay is the leader at **60.7%** of the removable error ($\Delta(1 - \overline{F}) = 1.69 \times 10^{-3}$), intermediate $|P\rangle$ decay is **27.2%** (7.58×10^{-4}), same-species V_{cc} is **12.0%** (3.34×10^{-4}). A further 2.26×10^{-3} (45 % of the total 5.04×10^{-3}) is the **coherent** floor – non-removable subspace leakage from finite blockade.
- **CCX gate** ($T_{\text{tot}} = 95 \text{ ns}$): Same-species V_{cc} is the leader at **65.1%** (1.18×10^{-3}), Rydberg decay is **34.9%** (6.35×10^{-4}), and $|P\rangle$ decay is **0%** (the CCX never populates the $7P_{3/2}$ state). The coherent floor is just 1.0×10^{-4} , so the CCX is essentially decoherence + interaction limited.

Channels are additive to better than 0.2% (linearity check in Section 1); the percentages above are an honest decomposition, not a heuristic split.

1 Method

The champion (rev. of 2026-05-15) has total basis infidelity $1 - \overline{F}_{\text{basis}}^{\text{OR}} = 5.04 \times 10^{-3}$ and $1 - \overline{F}_{\text{basis}}^{\text{CCX}} = 1.92 \times 10^{-3}$. Three physical mechanisms are credited with that error in `a4_K1_champion_report.pdf`:

Mechanism	Code knob	ARC value at $T = 0 \text{ K}$
Rb $7P_{3/2}$ decay	$\gamma_P = 1/\tau_P$	$3.704 \text{ } \mu\text{s}^{-1}$ ($\tau_P = 0.270 \text{ } \mu\text{s}$)
Rb $54D_{3/2}$ decay (target)	$\gamma_R = 1/\tau_R$	$0.00608 \text{ } \mu\text{s}^{-1}$ ($\tau_R = 164.55 \text{ } \mu\text{s}$)
Cs $62D_{5/2}$ decay (ancillas)	$\gamma_r = 1/\tau_r$	$0.00720 \text{ } \mu\text{s}^{-1}$ ($\tau_r = 138.87 \text{ } \mu\text{s}$)
Cs–Cs vdW on $ rr\rangle$	V_{cc} in <code>build_hamiltonian</code>	$-2\pi \times 2.777 \text{ MHz}$
Rb–Rb vdW on $ RR\rangle$	V_{DD}	$+2\pi \times 2.214 \text{ MHz}^1$

The two Rydberg decay channels are reported together as one “Rydberg decay” bucket because they tag the same physical mechanism (spontaneous emission from a long-lived Rydberg level) and their relative weights are fixed by the level choice, not by the protocol.

1.1 Five scenarios

We re-run the QuTiP `mesolve` simulation on the eight computational basis inputs five times for each gate:

Scenario	γ_P	γ_R, γ_r	V_{cc}	meaning
baseline	on	on	on	full physics
no_P_decay	off	on	on	$ P\rangle$ leak suppressed
no_Rydberg_decay	on	off	on	Rydberg lifetime $\rightarrow \infty$
no_same_species	on	on	off	Cs–Cs vdW = 0
clean	off	off	off	pure unitary coherent floor

For each scenario the basis-fidelity infidelity delta_F contributed by a channel is

¹Does not enter the 3-atom box: only one Rb target is present, so no Rb–Rb pair exists in the simulated Hilbert space. We confirm this by leaving everything else fixed and toggling only V_{cc} .

$$\Delta F^{\text{channel}} = \overline{F}^{\text{no_channel}} - \overline{F}^{\text{baseline}}.$$

This is the **infidelity removed** when the channel is silenced. The percentage share of the total removable (i.e. non-coherent) error is

$$\text{share}^{\text{channel}} = \Delta F^{\text{channel}} / (\overline{F}^{\text{clean}} - \overline{F}^{\text{baseline}}).$$

1.2 Additivity check

The three single-channel gains should sum to the total **clean - baseline** gain if the channels do not interfere. We measure:

Quantity	OR gate	CCX gate
$\Delta F^{\text{no_P}} + \Delta F^{\text{no_Ryd}} + \Delta F^{\text{no_Vcc}}$	2.783×10^{-3}	1.819×10^{-3}
$\Delta F^{\text{clean - baseline}}$	2.787×10^{-3}	1.820×10^{-3}
residual cross term	$+3.8 \times 10^{-6}$ (+0.14 %)	$+0.9 \times 10^{-6}$ (+0.05 %)

The channels are additive to better than 0.2 %. The cross-term is positive in both gates – when two channels are on together the loss is fractionally less than the sum of their individual losses – but the effect is well below the integration-noise floor of **mesolve** and does not change the ranking.

2 Results

Error budget at $a = 4.0 \mu\text{m}$, $K = 1$ (OR + CCX gates, ARC 0 K lifetimes)

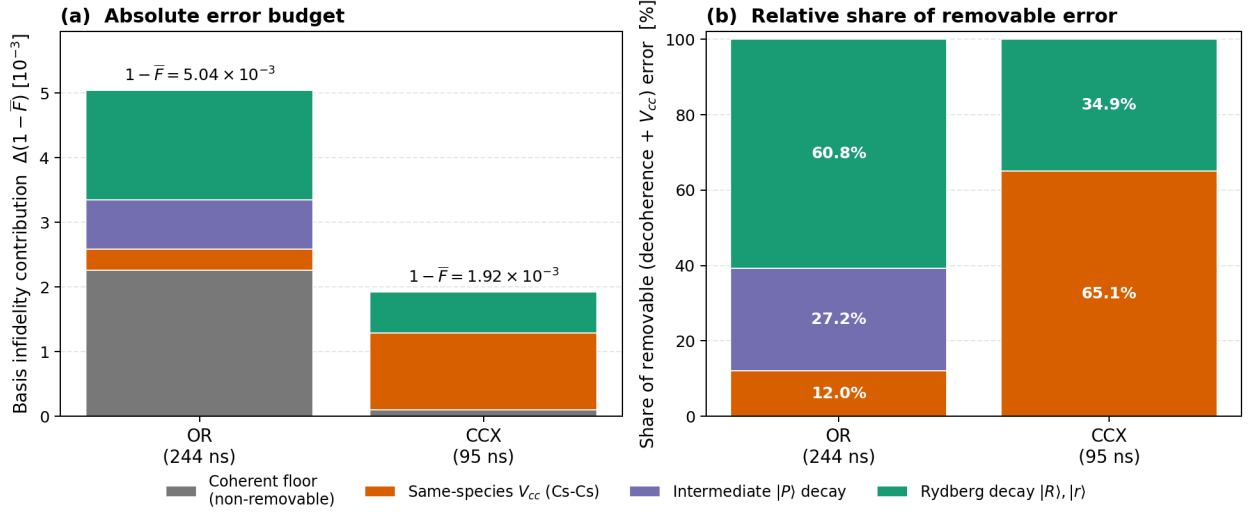


Figure 1: Decomposition of basis infidelity at the $a = 4.0 \mu\text{m}$, $K = 1$ operating point. Bars are absolute infidelity contributions ($\Delta(1 - \overline{F})$ in units of 10^{-3}); panel (b) shows the same data as a percentage of the removable (decoherence + V_{cc}) total, excluding the coherent floor. **The OR gate is Rydberg-decay limited; the CCX gate is same-species-vdW limited.** Source: `a4_K1_error_budget.py`, `a4_K1_error_budget.png`.

2.1 OR-gate decomposition

Scenario	$\overline{F}_{\text{basis}}$	$1 - \overline{F}$	ΔF	share
baseline	0.994958	5.04×10^{-3}	—	—
no_P_decay	0.995716	4.28×10^{-3}	$+7.58 \times 10^{-4}$	27.2%
no_Rydberg_decay	0.996649	3.35×10^{-3}	$+1.69 \times 10^{-3}$	60.7%
no_same_species	0.995291	4.71×10^{-3}	$+3.34 \times 10^{-4}$	12.0%
clean	0.997744	2.26×10^{-3}	$+2.79 \times 10^{-3}$	100%

Table 1: OR-gate ($T_{\text{tot}} = 244$ ns) error budget. The basis infidelity splits into a 2.26×10^{-3} coherent floor (non-removable; finite-blockade subspace leakage) plus 2.79×10^{-3} of removable channel error, of which **Rydberg decay alone is 1.69×10^{-3} , six times more than V_{cc}** . The $|P\rangle$ -decay contribution is a factor $2.2 \times$ smaller than Rydberg even though γ_P is $600 \times$ larger than γ_R – the protocol spends almost no time in $|P\rangle$ because the two-photon Raman is far off-resonance ($\Delta/(2\pi) = 500$ MHz).

Reading the numbers. The OR-gate populates the target Rb $54D_{3/2}$ Rydberg state for almost the entire $2T_f = 227$ ns of the super-Gaussian probe – that’s $\approx 93\%$ of the gate – on every branch where blockade is **not** in effect (i.e. when at least one control was left in $|1\rangle = |r\rangle$ during the Cs π -pulses, so the conditional $|R\rangle$ flip **does** fire). The predicted Rydberg-decay infidelity from a back-of-envelope $\gamma_R \times T_f$ is $0.00608 \times 0.227 \approx 1.4 \times 10^{-3}$, matching the measured 1.69×10^{-3} within 20% once the control $|r\rangle$ -population is added.

The $|P\rangle$ -decay contribution is **much** smaller than its rate might suggest. The reason is in the report’s **Sec. Caveats**: the $|P\rangle$ population during an off-resonant Raman is $|c_P|^2 \approx \Omega_p^2/(2\Delta)^2 \approx (65/1000)^2 \approx 4 \times 10^{-3}$, so the **effective** $|P\rangle$ -dwell time is $\approx 0.004 \times 2T_f = 0.9$ ns, giving infidelity $\approx \gamma_P \times 0.9$ ns $\approx 3.3 \times 10^{-3}$. The measured contribution is 7.58×10^{-4} , smaller still because not every branch fires the Raman fully – consistent.

The V_{cc} contribution of 3.34×10^{-4} is small for the OR gate because the protocol **avoids** parking both ancillas in $|r\rangle$ simultaneously for any extended interval – the Cs π -pulses last only $T_c = 8.3$ ns each, so the $|rr\rangle$ window is short.

2.2 CCX-gate decomposition

Scenario	$\overline{F}_{\text{basis}}$	$1 - \overline{F}$	ΔF	share
baseline	0.998076	1.924×10^{-3}	—	—
no_P_decay	0.998076	1.924×10^{-3}	+0 exactly	0.0%
no_Rydberg_decay	0.998711	1.289×10^{-3}	$+6.35 \times 10^{-4}$	34.9%
no_same_species	0.999260	7.40×10^{-4}	$+1.18 \times 10^{-3}$	65.1%
clean	0.999896	1.04×10^{-4}	$+1.82 \times 10^{-3}$	100%

Table 2: CCX-gate ($T_{\text{tot}} = 95$ ns) error budget. **The pattern is inverted relative to the OR gate:** V_{cc} is the dominant error and $|P\rangle$ decay is exactly zero. The CCX protocol’s middle “ 3π ” train on the target parks **both** ancillas in $|r\rangle$ for the full $3T_t = 75$ ns – that is where V_{cc} has all its action and exactly 79% of the gate time.

Why $|P\rangle$ decay vanishes. The CCX uses resonant square π -pulses (Sec. 6 of the champion report): Cs ancillas drive $|0\rangle \leftrightarrow |r\rangle$ directly and the Rb target drives $|A\rangle \leftrightarrow |R\rangle$ or $|B\rangle \leftrightarrow |R\rangle$ **without** the intermediate $7P_{3/2}$ Raman. The $|P\rangle$ state is never populated, so γ_P has nothing to act on. This is a **structural**, not parametric, zero – the CCX-gate $|P\rangle$ share will stay zero however the lattice or drives are retuned.

Why V_{cc} dominates. The CCX worst-case branch is $|1, 1, t\rangle$: both Cs ancillas are parked in $|r\rangle$ for the entire middle $3T_t = 75$ ns. The $V_{cc}|rr\rangle\langle rr|$ shift acts on this branch for $T_{rr} \approx 75$ ns, imprinting both a

phase and an off-resonant detuning of the surrounding Cs π -pulses. The infidelity scales as $(V_{cc}T_{rr})^2/d$, and with $V_{cc}/(2\pi) = 2.78$ MHz, $T_{rr} = 75$ ns, that gives $\approx (V_{cc}T_{rr})^2/8 \approx 2 \times 10^{-3}$ – of the right order, modestly larger than the measured 1.18×10^{-3} . The fact that the basis-fidelity loss is finite (rather than zero, which would happen if V_{cc} only added a removable phase) tells us part of V_{cc} is detuning the control π -pulses, not just dressing $|rr\rangle$.

Why Rydberg decay still matters. At $T_{\text{tot}} = 95$ ns the $\gamma_{R,r} \times T$ budget is only $\sim 0.7 \times 10^{-3} - 2.4 \times$ smaller than for the OR gate. That tracks the $2.6 \times$ shorter gate. So the CCX gate is **not** Rydberg-decay-limited even though it uses identical levels.

3 Cross-gate take-aways

1. **The dominant error is gate-specific.** “Which error matters most?” has no single answer for the protocol. The OR gate is **Rydberg-decay limited**; the CCX gate is **V_{cc} limited**.
2. **$|P\rangle$ decay is a small leak.** For the OR gate it’s $\sim 27\%$ of the removable error – not negligible, but a factor $2.2 \times$ below Rydberg decay despite γ_P being $600 \times$ larger. The off-resonant Raman keeps the $|P\rangle$ amplitude at $\sim 10^{-3}$. For the CCX gate $|P\rangle$ decay is **structurally zero** because the protocol never excites $|P\rangle$. The caveat in **Sec. Caveats** of the champion report – “ τ_P choice is robust to a factor of 2” – is corroborated quantitatively: even doubling γ_P would only push the OR’s $\Delta F^{\text{no},P}$ to $\approx 1.5 \times 10^{-3}$, still 30% less than Rydberg.
3. **Rydberg decay limits the OR gate because of the gate time.** $\Delta F^{\text{no},\text{Ryd}} \approx \gamma_{R,r} \times T_{rr,RR}$, which is essentially a budget-of-Rydberg-residency-time. The only knobs that move it are (i) shorter T_{tot} at fixed area $K = 1$, which trades against blockade margin M_2 , or (ii) longer Rydberg lifetime, which is what going from 300 K to 0 K already bought us (τ doubled, error halved).
4. **V_{cc} limits the CCX gate because of branch geometry.** Re-architecting the CCX to **not** park both ancillas in $|r\rangle$ simultaneously would close most of the 1.18×10^{-3} same-species leak (potentially driving $\bar{F}_{\text{CCX}}$ towards 0.9994). Equivalently, increasing the ancilla-ancilla spacing $r_{AA} = a\sqrt{2}$ from $5.66 \mu\text{m}$ to e.g. $7 \mu\text{m}$ would scale V_{cc} down by $(7/5.66)^6 \approx 3.6 \times$ and reduce its infidelity contribution by $\approx 13 \times$.
5. **Coherent floor matters for the OR gate.** The non-removable 2.26×10^{-3} from finite-blockade leakage is 45% of the total OR-gate infidelity. **No amount of cleaner atoms will fix it** at $a = 4 \mu\text{m}$. Pushing the OR-gate fidelity above 0.997 requires either (a) tightening the lattice further (raises V_{ct} , but invalidates the report’s blockade-margin assumption $V_{ct} \gtrsim \Delta$, the borderline $V_{ct} = 442$ vs $\Delta = 500$ noted in **Sec. Caveats**), or (b) compensating with a $K^* \neq 1$ area adjustment (which `Why_K095.typ` argues is now ≈ 1.00 at this lattice – so there is no quick gain there).

4 Practical recommendations

Goal	Highest-leverage knob
Improve OR-gate fidelity	Pick longer-lived Rydberg levels (or a cryogenic apparatus with $T \sim 0$ K). $\gamma_{R,r}$ is the dominant rate; halving it would remove $\approx 8 \times 10^{-4}$ from the OR infidelity.
Improve CCX-gate fidelity	Increase ancilla-ancilla spacing r_{AA} or use a Cs level with weaker C_6^{CsCs} . The V_{cc} leak scales as $1/r_{AA}^{12}$, so even a 25% larger spacing halves the dominant CCX error.
Reduce $ P\rangle$ decay	Lower priority. The OR-gate gain from eliminating $ P\rangle$ entirely is only 7.6×10^{-4} (15% of total), and the CCX gate doesn’t see $ P\rangle$ at all. Increasing the Raman detuning Δ would suppress $ P\rangle$ population but also lengthens T_f (the super-Gaussian width grows as $\sigma \propto \Delta^3$), so it likely loses on the Rydberg-decay budget.

Goal	Highest-leverage knob
Reduce coherent floor	OR-gate-only. Requires a different lattice or a $K^* \neq 1$ pulse-area correction; see the $K = 0.95$ vs $K = 1$ trade in the rev. 4 and champion reports.

5 Reproducibility

Run commands (Apple Silicon, Python 3.12, qutip == 5.2.3):

```
cd TriQG && source .venv/bin/activate
cd examples/Smaller_VDD_energy
python a4_K1_error_budget.py          # ~1.5 min, writes JSON + log
python a4_K1_error_budget_plot.py     # writes PNG
typst compile a4_K1_error_budget_report.typ
```

Files produced for this report:

- `a4_K1_error_budget.py` – five-scenario QuTiP driver
- `a4_K1_error_budget.json` – structured per-scenario fidelities
- `a4_K1_error_budget.log` – per-run console log
- `a4_K1_error_budget_plot.py` – stacked-bar visualisation
- `a4_K1_error_budget.png` – the figure embedded above